EconDID Studio

Guided, auditable DID and DDD analysis for applied researchers

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Repository: https://github.com/panzehui-gh/econ-did-studio

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Purpose and user workflow

EconDID Studio reduces the repeated programming work between a research design and an estimable difference-in-differences (DID) or triple-differences (DDD) model. A researcher uploads a CSV, TSV, or Excel file, selects the closest model preset, confirms the meaning and positive direction of variables, optionally adds controls/fixed effects, and downloads a reproducibility bundle. The main interface is a Streamlit web app, while a command-line interface supports batch and scripted use. The current scope contains basic 2×2 DID, panel DID with entity/time fixed effects, custom-fixed-effect DID, and DDD.

Automation does not replace identification. The interface explicitly warns that causal interpretation requires assumptions such as parallel trends and that its trend plot is descriptive. Ambiguous two-category labels are never silently assigned to treatment or control.

Implementation and design choices

The pipeline is non-destructive and records every operation: label normalization, empty/duplicate removal, missing-marker standardization, and conservative numeric/date conversion. An explainable heuristic ranks candidate columns for outcome, treatment, post, entity, time, and subgroup. It combines multilingual name aliases, fuzzy similarity, data type, cardinality, and optional plain-language descriptions. The user sees confidence, reasons, alternatives, and confirms the final mapping.

Models are built with statsmodels formulas. HC1 covariance is the default; cluster-robust covariance is available and is selected automatically by entity for the two-way fixed-effect preset. Pre-estimation checks require all four DID cells (or eight DDD cells), exactly two indicator values, complete estimable observations, and the required panel identifiers. The export ZIP contains the exact analysis data, coefficient table, formula/coding metadata, diagnostics, full statsmodels summary, and observed-trend figure. The package also provides synthetic data generators, a CLI, unit tests, MIT license, and GitHub Actions CI.

Objective validation

Synthetic balanced panels were generated from known data-generating processes. Twenty independent seeds were used for each estimator (40 fits total), with entity-clustered standard errors and entity/time fixed effects where appropriate. For DID, the true effect was 2.0: mean estimate 1.9988, bias −0.0012, RMSE 0.0904, and 95% interval coverage 0.95. For DDD, the true effect was 1.5: mean estimate 1.4512, bias −0.0488, RMSE 0.1496, and coverage 1.00. The automated test suite also checks dirty-data conversion, multilingual role mapping, indicator-direction safeguards, missing-cell rejection, input immutability, result export, and a headless run of the web app.

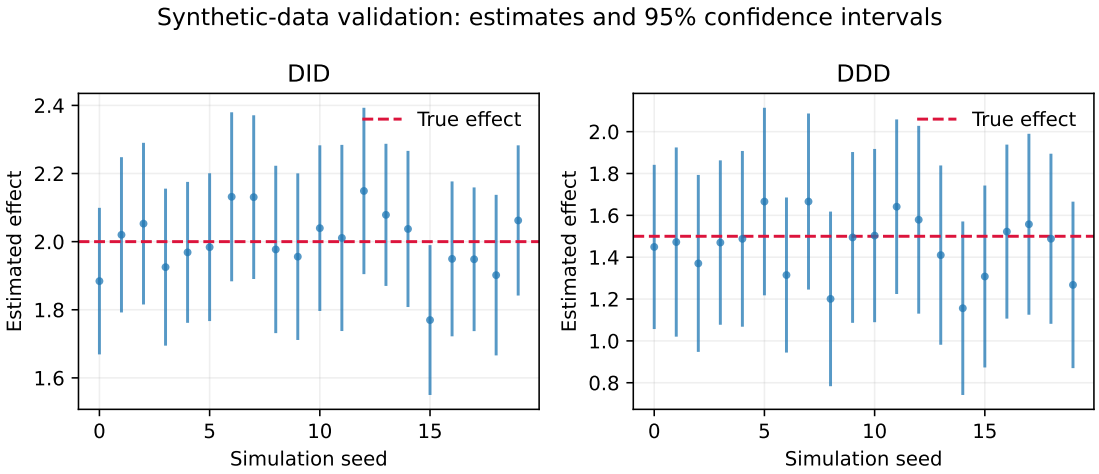

Figure 1: Estimates and 95% confidence intervals across synthetic seeds; the dashed line is the known effect.

AI use and reproducibility

OpenAI Codex (GPT-5) was used to help translate requirements into a software architecture, draft implementation/tests/documentation, and diagnose test failures. I reviewed the generated code through executable unit tests and known-effect simulations rather than accepting output by inspection. No secret, personal, or externally supplied research data is included. Dependency versions are declared in `pyproject.toml`; installation, launch, CLI examples, test commands, limitations, and CI status are documented in the repository README.